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28 November 1972
SUPERSEDING
DAPD-829
7 October 1963

MILITARY SPECIFICATION

WELDING, FLASH, STANDARD LOW CARBON STEEL

1. SCOPE

1.1 Scope. This specification covers procedures for flash welding standard low carbon steels, with a carbon content of 0.23 maximum.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification.

STANDARDS
Federal
Fed. Test Method Std. No. 151 - Metals:
Test Methods.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. EQUIPMENT

3.1 Requirements. Means shall be provided for automatically controlling the feed of the joint during flashing, the rates and distance of travel of the sections to be welded, and for controlling the secondary voltage and current magnitudes and timing, and the time of current cutoff.

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3.1.1 The equipment shall be capable of bringing the two sections to be welded into contact, initiating the flow of welding current and forming a "flashing" action between the adjacent pieces of metal. As the metal on the surfaces is burned or flashed away, the movable platen shall advance and maintain the gap within appropriate limits such that flashing is maintained until the mating surfaced and adjacent areas have attained a degree of plasticity suitable for welding. The surfaces shall be rapidly brought together with sufficient force to close all craters formed during flashing, expel molten metal and substantially all oxides and impurities from the weld proper, upset the adjacent metal, and produce welds of acceptable quality.

3.2 Operation. Once the process has been initiated by the operator, the equipment shall follow the sequence of operations automatically and complete the weld. The equipment shall consistently produce welds which meet the quality standards specified herein. Manual flashing not to exceed 15 percent of the total flashing travel will be permitted.

3.3 Electrodes. The electrodes shall be of adequate size, and shall be so fitted to the surfaces of the parts to be welded that the necessary current will be uniformly distributed about the contact zone, and shall be conducted from the electrode to the work without damage owing to localized overheating.

3.4 Control of machine variables. The equipment shall be set up by experienced operators under the supervision of a welding engineer or other authorized personnel.

3.4.1 Welding schedules. Complete flash welding procedure schedules shall be established as specified herein for the joints to be welded in production and approved by the Government Inspector before a machine may be used for the welding of items to be furnished on Government contract.

4. MATERIALS AND PREPARATION

4.1 Materials. Standard low carbon steels of not more than 0.23 maximum carbon content may be welded provided that all requirements of this specification are met.

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4.1.1 Cleaning. The surfaces of the parts contacting the electrodes, and the surfaces at the fusion zones, shall be clean and free from oxides, paint, grease, dirt, or foreign matter which would interfere with the flow of current or fusion of the metal. Chemical cleaning or grit blasting are preferred. Sand blasting may be employed provided the surfaces are treated to remove imbedded sand particles prior to welding.

4.1.2 Joint preparation. The mating parts shall be alined in such a manner that the heat generated by the flow of current will be uniformly distributed over the section. The details of joint preparation shall be determined by the contractor and shall become a part of the welding schedule for the respective joint. Recommended pratices are shown in Figure 1.

4.1.3 Limitation on tubing. Tubing possessing a ratio of outside diameter to wall thickness of greater than 30:1 shall not be flash-welded unless the equipment is capable of meeting the more rigid alinement tolerances for 35:1 and 40:1 ratios specified in 5.2.1.

5. QUALITY STANDARDS

5.1 Flash-welded joints shall be sound and free from injurious defects, such as cracks, inclusions, die burns, and porosity which affect the strength or serviceability of the part.

5.1.1 Flash removal. A sufficient amount of the extruded or upset metal shall be removed from all weld areas in order that no defective or unsound material remains. A weld shall be rejected if it is necessary to machine below the level of the surface of the parent metal to remove surface defects. Unless otherwise specified, the removal of the internal flash from joints in hollow shaped will not be required.

5.2 Alinement tolerances.

5.2.1 Sheet and tubular joints. For sheet thickness or wall thickness of tubing 0.080 inch and less, the misalinement of thickness shall not exceed 0.008 inch. For sheet or wall thickness greater than 0.080 inch, the misalinement of thickness shall not exceed 10 percent of the nominal section thickness up to a maximum of 0.030 inch, except for the following two higher ratios where a smaller misalinement is required:

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O. D. Wall Ratio (maximum)	Misalinement Tolerance Percent Maximum
35:1	6
40:1	4

Misalinement shall be measured at, or near, the plane of the weld before the flash is removed.

5.2.2 Bars. The misalinement of the surfaces in joints between bars and rods shall not exceed 5 percent of the diameter or thickness when measured at, or near, the plane of the weld, before the flash is removed.

5.2.3 Angular misalinement. The angular misalinement of the welded joints shall not exceed 0.005 inch per inch of length.

6. CERTIFICATION OF WELDING SCHEDULES

6.1 General. The welding schedule shall be established by the contractor for each joint differing in material, area, or contour from joints previously welded in accordance with approved schedules, or for each welding machine over an appropriate range of cross-sectional areas, or wall thickness, or both.

6.2 Welding schedule. A complete welding schedule shall include the preparation of the parts to be welded, choice of electrodes and fittings, and appropriate machine control devices and settings to provide the proper sequence of events. The record of a welding schedule shall include the appropriate items of Figure 2.

6.3 Establishing welding schedules.

6.3.1 General. The three initial welding schedules established by the contractor for a given machine shall represent joints of the greatest and least cross-sectional areas to be welded in production plus one joint of intermediate area. To determine the acceptability of the settings and the consistency of the resulting welds, fifteen or more specimens shall be welded at each of the three points, the welds tested to destruction in tension or bending, and the test data submitted to the authorized Inspector with a request for approval of the welding schedules represented. Assemblies welded with the

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respective machine shall not be acceptable as items furnished on a Government contract until approval of the welding schedules has been obtained. Weld schedules shall not be changed during a production run, except as provided in 6.3.2.

6.3.2 Additional schedules or modification of existing schedules. To weld parts with cross-sectional areas other than those specifically checked above, or to accomplish minor modification of existing schedules as may be necessary, the contractor shall demonstrate that the settings for these secondary points are acceptable on five sample parts or control specimens. If unsatisfactory results are obtained for a secondary point, a sample of 15 shall be run to check the point, after suitable adjustments in settings have been made. Specific approval of each additional schedule will not be required, but complete records shall be made readily available to operating personnel and authorized inspection personnel for reference at any time.

6.3.3 Production run. Unless otherwise specified, one setting of the machine shall satisfactorily weld and maximum and minimum cross-sectional areas scheduled during a production run. No change in setting shall be made during a run without the approval of the authorized Inspector.

6.3.4 Individual part certification. When one, or a limited number of joint types are to be welded in production and complete qualification of the machine is not considered necessary or desirable, the contractor may establish individual schedules on the basis of 15 test specimens, subject to approval by the authorized Inspector.

6.4 Certification test specimens. Either production parts or simulated parts may be used in welding certification test specimens. The parts shall be prepared under production condition. The mating surfaces to be joined and the electrode contact areas shall be identical with the production part in cross-sectional area, contour, and finish. The joints shall be welded in accordance with the proposed welding procedure and with the requirements specified herein.

6.4.1 Type of specimens. Specimens shall be tested in full section whenever practicable. Where test specimens are too large to be tested in full section, tensile coupons conforming to an appropriate type, as defined under Method 211 of Federal Standard No. 151 may be cut from locations, such that the weld zone is included in the test

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section. The aggregate weld area of the coupons shall be approximately 50 percent of the area of the weld joint represented.

7. TEST PROCEDURES

7.1 Contractor inspection. Unless otherwise specified herein, the inspections required by this specification shall be performed by the contractor, under the surveillance of the cognizant Government Inspector.

7.2 Production control. All specimens required herein for checking production flash welding shall be welded in accordance with the established schedule.

7.3 Sampling.

7.3.1 Sample testing. Unless otherwise specified on the drawing, in the contract, or order, the amount and type of testing other than visual, shall be performed at the discretion of the Government Inspector.

7.4 Test methods.

7.4.1 Visual examination. Each flash-welded joint shall be examined for cracks, die burns, telescoping, and other indications of unsound welding before and after the flash is removed.

7.4.2 Destructive tensile test. Destructive tensile test shall be performed in accordance with the requirement specified in 6.4. Other methods of destructive tests such as bend tests, impact tests or short-time fatigue tests, which are sensitive to variation in weld quality may be used when approved by the procuring activity.

7.4.3 Tensile tests. The whole weld test specimen shall be subjected to tensile tests in accordance with Federal Test Method Standard No. 151. Where test specimens are too large to be tested in whole, they may be cut into tensile coupons prior to testing in accordance with Federal Test Method Standard No. 151, except that reduction in size of the test section of the specimens will not be required. The entire test specimen, except that removed by machining, shall be tested when cut into coupons. Unless at least three fractures

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occur in the parent metal, at least three additional tests shall be made on the latter, by retesting a (tubular) part of the test specimen or by testing coupons cut from the specimen.

7.5 Test requirements.

7.5.1 Whole specimens. The results of the tests on whole specimens shall conform to the following:

(a) No specimen tested in the "as welded" condition shall fail in the weld.
(b) For other specimens and conditions, failure in the weld shall be acceptable provided the stress at failure is not less than 95 percent of the ultimate strength of the base metal or 95 percent of the average ultimate strength of those specimens (if two or more in number) in which base metal failures occurred. The strength of the base metal shall be determined by retesting a part of the original specimen or by testing a standard tensile coupon machined therefrom.

7.5.2 Strip test coupons. Provided that no defects associated with faulty welding conditions are present, the results of tests on coupons shall be acceptable when:

(a) The average strength of all the coupons from each specimen is not less than 95 percent of the average strength of the parent material.
(b) The minimum test value shall be not less than 80 percent of the average strength of all the coupons, except that one coupon may have a test value below 80 percent but above 60 percent, provided the previous history of the tests for the particular setting shows that the frequency of occurrence of the lower test values is not greater than 1 percent.

7.5.3 Machine maintenance. The contractor shall satisfy the Inspector that all resistance welding equipment receives adequate maintenance attention, that the operation of all component parts is

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checked periodically, and that any defective parts are replaced before operation is resumed.

7.5.4 Retests. If the results of tests for a particular machine setup are not satisfactory, the machine setup shall be checked and any unsatisfactory condition corrected. The machine setup shall then be recertified using five specimens in accordance with the requirements of 6.4.

7.5.5 Records. The contractor shall keep a record of all tests and a list of all parts corresponding to each setup and test. The record shall include also the machine settings and other data. The records shall be subject to examination by the Inspector.

8. NOTES

8.1 Intended use. The procedures covered by this specification are intended for Military tracked and wheeled vehicle applications and other applications where control procedures and desired.

8.2 Definitions. The definitions of the terms herein are, in general, in accordance with American Welding Society's recommened practice for the flash welding of low and medium forging strength steels.

8.2.1 "Weld failure" as used herein is a failure either partially or wholly in the weld plane except for partial weld-plane failures in heat-treated whole specimens. Partial weld-plane failure in heat-treated whole specimens is not considered a weld failure where fracture in the weld plane does not exceed 25 percent of the weld area and is accompanied by visual necking down of the parent metal.

8.2.2 A "production run" as referred to herein is the production flash welding of identical parts without a change in machine setup for 4 hours of continuous operation or for a period of time acceptable to the Inspector.

8.3 Protective atmosphere welding. For greater assurance of weld quality, tubing with wall thickness over 0.149 inch should be welded in a protective atmosphere.

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8.4 Figure 3, 4, and 5 are examples of initial curves established for a machine.

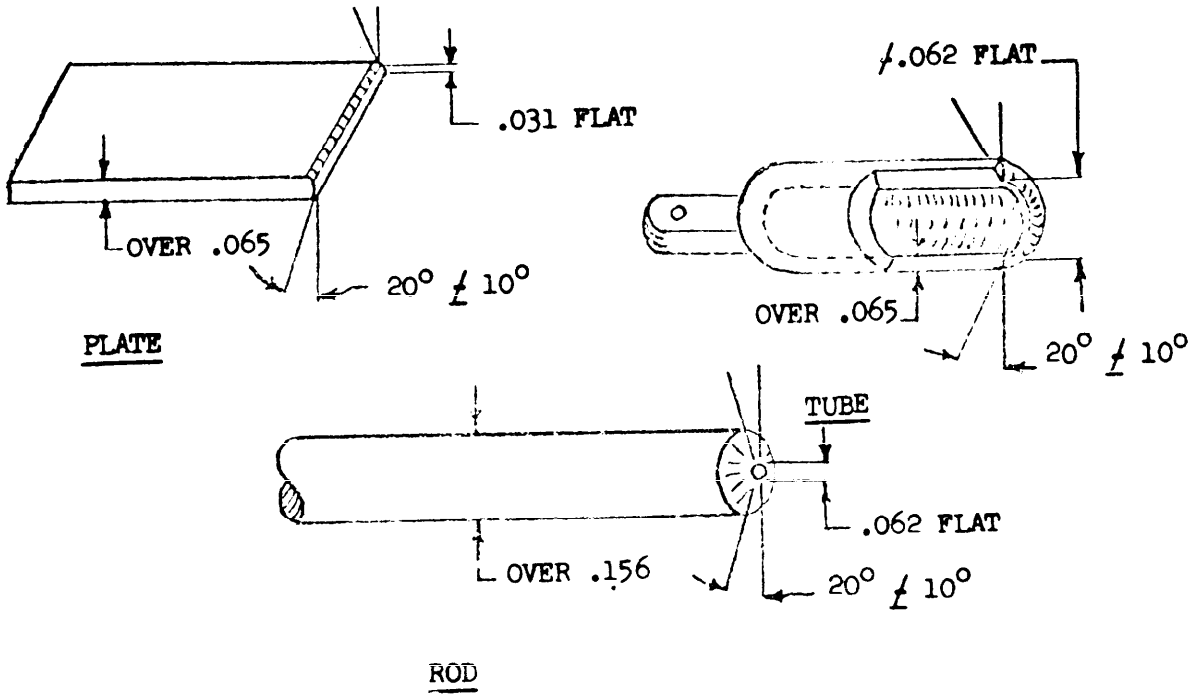
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NOTE: a. THE ENDS OF PARTS TO BE WELDED SHALL BE OUT SQUARE WITHIN PLUS OR MINUS 1/2 A DEGREE.

b. WHEN THE NOMINAL STOCK THICKNESS EXCEEDS THE VALUES GIVEN IN THIS FIGURE, CHAMFER ONE OF THE PARTS AS SHOWN. IN THE EVENT IDENTIDAL PARTS ARE TO BE WELDED TO EACH OTHER RESULTING IN EACH BEING CHAMFERED, CHANGE THE ANGLE TO 10°.

UNLESS OTHERWISE SPECIFIED, DIMENSIONS IN INCHES.

FIGURE 1. Recommended end preparation for flash butt welding of tubing, flat sheets, and bar.

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Manufacturer of machine_____
Type and serial number_____
Capacity of machine_____
Transformer tap setting_____
Transformer turns ratio_____
Secondary voltage (Approx)_____
Flashing time (seconds)_____
Beginning of upset to current cutoff_____
Burnoff (inches)_____
Upset travel (inches)_____
Platen travel control cam (when used)_____
Area of electrode contact areas_____
Work locaters employed_____
Material and surface condition_____
Details of joint preparation: (Show sketch and dimensions.)

_____
*Signature (Welding Engr)

_____
*Signature (Govt Inspector)

*NOTE: The supervisory welding engineer or authorized alternates and the authorized Government Inspector shall sign schedule form.

FIGURE 2. Record of welding schedule.

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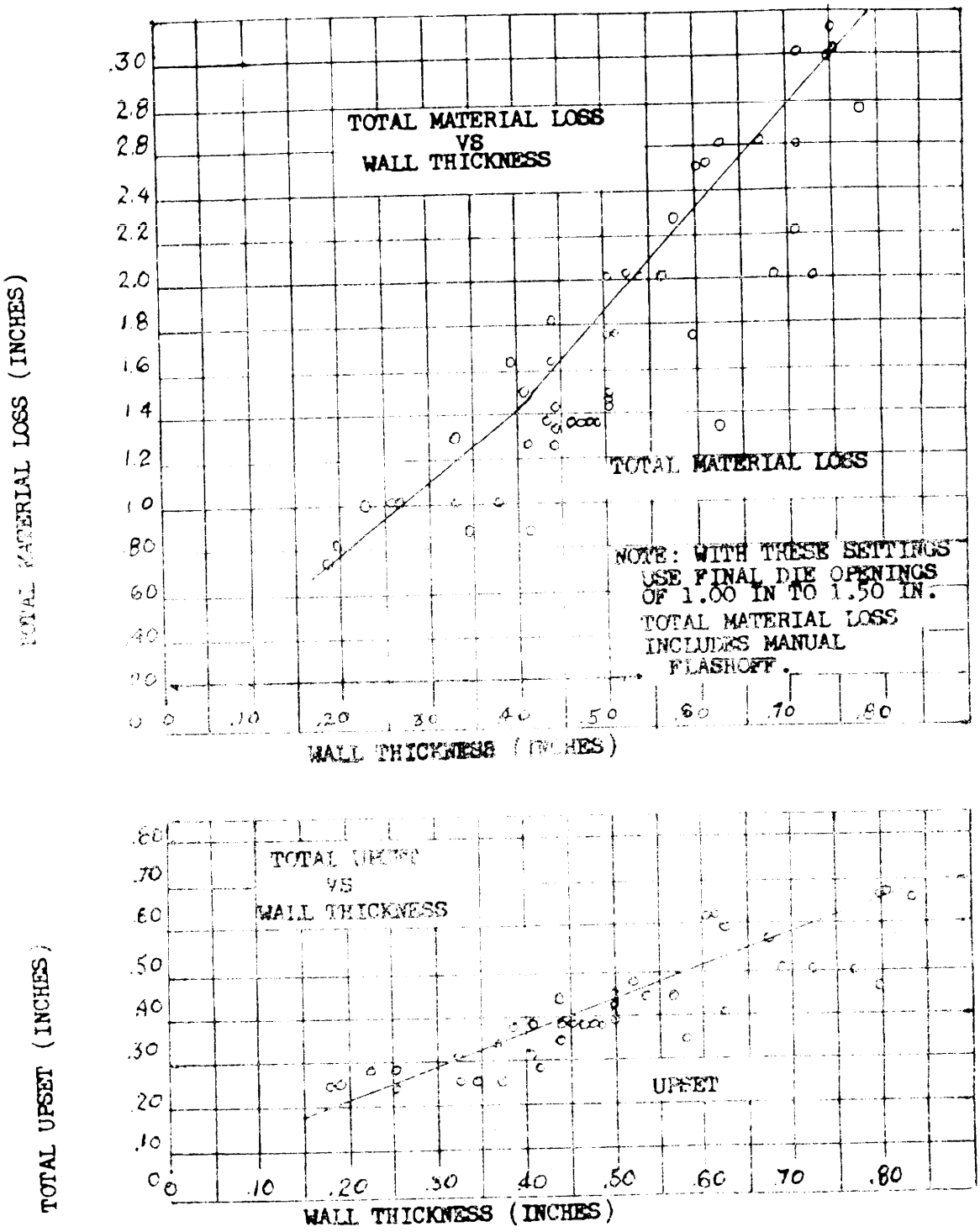


FIGURE 3. MACHINE SETTINGS VERSUS WALL THICKNESS (SPECIMEN GRAPH).

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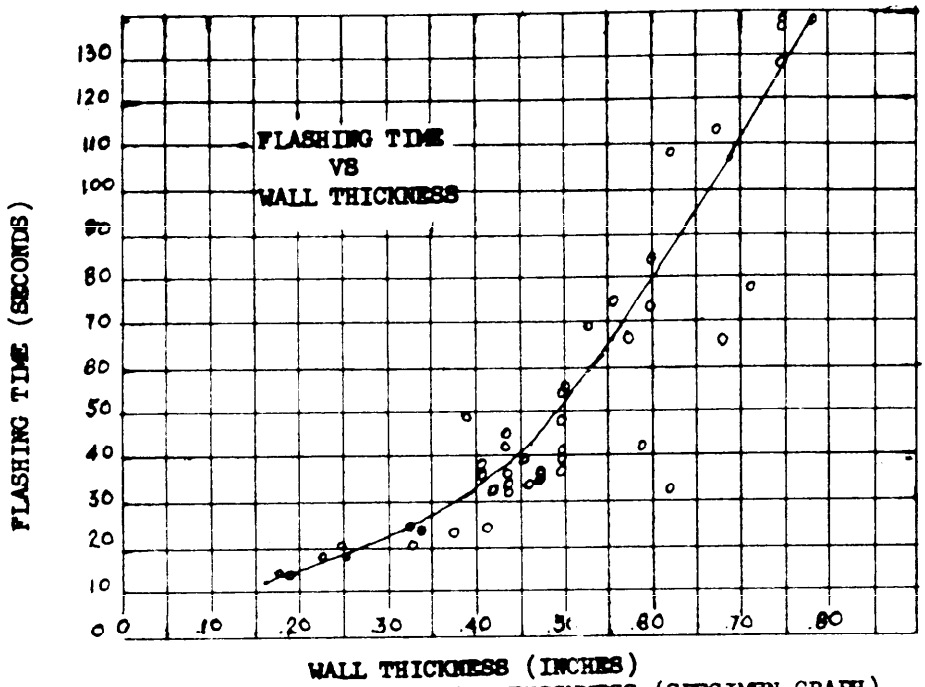


FIGURE 4. FLASHING TIME VERSUS WALL THICKNESS (SPECIMEN GRAPH)

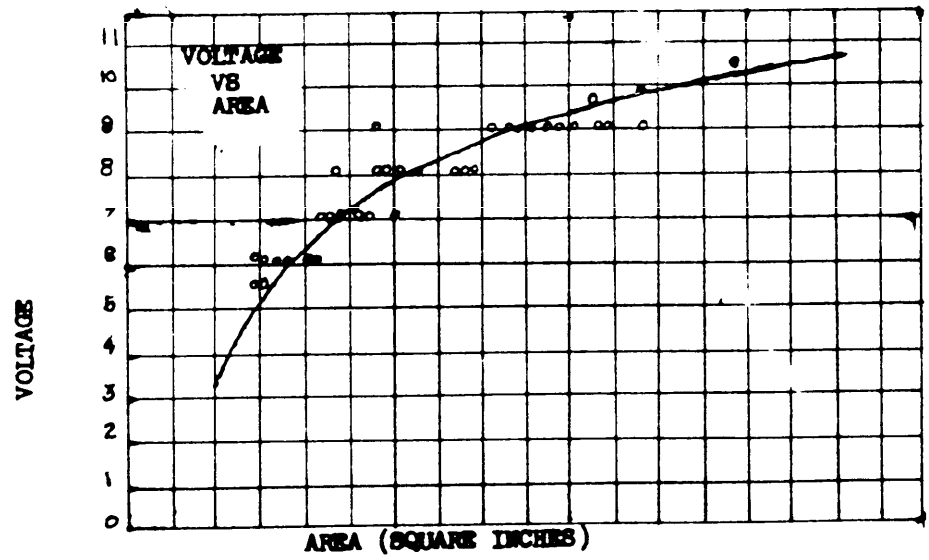


FIGURE 5. VOLTAGE VERSUS CROSS-SECTIONAL AREA (SPECIMEN GRAPH)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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